

UK Employment Insights

A comprehensive exploration using R programming, statistical methods, and data visualization.

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Introduction to the Domain

Uncovering Trends in the UK Job Market through Data Analytics

◆ Labor Market Insight Imperative:

- Understanding employment trends is crucial for job seekers, policymakers, and recruiters.
 - Job listing analytics helps reveal in-demand skills, competitive locations, and salary benchmarks.
 - Data-driven career planning improves alignment between workforce supply and demand.
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◆ Dataset Overview:

- Analysis of a dataset containing over 100,000 job listings from Reed.co.uk, a leading UK job portal.
 - Dataset includes key fields: job title, company, location, job type, salary, posting date, and description.
 - Rich textual descriptions allow natural language processing (NLP) for keyword insights.
-

◆ Data Characteristics:

- Data covers diverse industries and regions across the UK.
- Fields like job_title, category, salary_offered, and post_date enable structured and temporal analysis.
- Job descriptions vary in length and style, allowing exploration through tokenization and word clouds.

◆ Link to Dataset:

- (<https://www.kaggle.com/datasets/jobspikr/50000-job-board-record-from-reed-uk?resource=download>)

Project Objectives

Aims and Analytical Scope of the UK Jobs Data Project

Project Objectives

-  **In-depth Data Exploration:** Analyze job listings to uncover patterns in job titles, companies, categories, salary ranges, and posting trends across UK cities and regions.
 -  **Data Cleaning and Preparation:** Handle missing values, standardize formats, remove salary outliers, and prepare data for analysis and visualization.
 -  **Text Analysis:** Use Natural Language Processing (NLP) techniques to extract common terms and keywords from job descriptions.
 -  **Visual Insights:** Create clear and informative visualizations to understand top job roles, cities, categories, and salary distributions.
 -  **Interactive Job Search Tool:** Implement user-based filtering to search jobs by field and state directly within R.
-

Statistical and Exploratory Evaluation

- Analyze salary trends, frequency of job postings, and city-wise job availability.
 - Perform comparative analysis across job types and categories.
-

Hypotheses

- H0: There is no significant difference in job demand across categories and cities.
 - H1: There is a significant difference in job demand among categories and cities in the UK job market.
-

Installing Necessary Libraries and Packages

Setting Up the R Environment for Data Processing and Visualization

Purpose

- Ensure all required R packages are installed and loaded before analysis.
 - Enable essential functionalities like data wrangling, visualization, date handling, text mining, and LaTeX rendering.
-

Key Libraries and Their Roles

- **tidyverse** – Core library for data manipulation and visualization (includes dplyr, ggplot2, etc.)
- **lubridate** – Simplifies date parsing and manipulation
- **janitor** – Cleans column names and tidies messy data
- **skimr** – Provides summary statistics in a readable format
- **tidytext** – Powers text tokenization and frequency analysis
- **wordcloud** – Visual representation of most common words in job descriptions
- **RColorBrewer** – Enhances plot color palettes
- **stopwords** – Supplies lists of common stopwords for text cleaning
- **tinytex** – Installs and manages LaTeX environment for document output

```
install.packages("tidyverse")
install.packages("lubridate")
install.packages("janitor")
install.packages("skimr")
install.packages("tidytext")
install.packages("wordcloud")
install.packages("RColorBrewer")
install.packages("stopwords")
install.packages("tinytex")
tinytex::install_tinytex(force = TRUE)
```

```
library(tidyverse)
library(lubridate)
library(janitor)
library(skimr)
library(tidytext)
library(wordcloud)
library(RColorBrewer)
library(stopwords)
```

Investigating the Dataset

Loading and Exploring the Reed UK Job Listings Dataset

```
# =====  
# STEP 2: Load Data  
# =====  
job_in_uk <- read_csv("reed_uk.csv")  
# Replace with your file path
```

	category	city	company_name	geo	job_board
1	catering jobs	Chudleigh	Haulfryn Group	uk	reed
2	law jobs	Swindon	Reed	uk	reed
3	factory jobs	Norfolk	Swanstaff Recruitment Ltd	uk	reed
4	law jobs	Hurstpierpoint	Castles Solicitors	uk	reed
5	factory jobs	Hertfordshire	Marketplace Group	uk	reed
6	retail jobs	Belfast	Vodafone	uk	reed
7	factory jobs	Thetford	Hales Group Limited	uk	reed
8	motoring automotive jobs	Luton	Fine Recruiting Ltd	uk	reed
9	motoring automotive jobs	Blackburn	Audi	uk	reed
10	law jobs	Leeds	Rachel Clark Legal Recruitment	uk	reed
11	motoring automotive jobs	Bermuda Park	Keastone Recruitment	uk	reed
12	law jobs	Leeds	IPS Group	uk	reed
13	retail jobs	Stevenage	Matalan	uk	reed
14	retail jobs	London	New Look Retailers Ltd	uk	reed

job_description	job_requirements	job_title
Apply now New opportunity not to be missed! Haulfryn Gro...	NA	Commis Cl
Apply now This role requires an experienced secretary with ...	NA	Corporate
Apply now Job Title: Warehouse personLocation: SohamRat...	NA	Warehouse
Apply now Castles Solicitors are looking for a part / full time...	Required skills Admin Case Management Correspondence L...	Legal Secre
Apply now International Pharmaceutical Manufacturer requi...	NA	Category P
Apply on employer's website Retail Advisor - Belfast Donne...	NA	Retail Advi
Apply now Break Press Operator - Permanent, Â£9.25 â€” Â£...	NA	Break Pres
Apply now An exciting opportunity to start with a new accid...	Required skills Accident Paint Repair Spray Paint	Vehicle Pai
Apply on employer's website This exciting opportunity exist...	NA	Vehicle Tec
Apply now This highly regarded, top law firm in the North o...	Required skills Diary Management Legal Secretarial Typing	LEGAL SEC
Apply now The Opportunity This is a remarkable opportunit...	Required skills Car Dealership Sales Sales Executive Sales Te...	Sales Execu
Apply now Talented junior solicitor is required by a leading ...	NA	Commerci
Apply on employer's website About the Role As an Assistant...	NA	Assistant S
Apply on employer's website Every person makes New Look...	NA	Buying, Me

job_type	post_date	salary_offered	state
Permanent, full-time	3/12/2018	£18,323 per annum	Devon
Permanent, full-time	3/8/2018	£18,000 - £22,000 per annum	Wiltshire
Permanent, full-time	2/27/2018	Salary not specified	East Anglia
Permanent, full-time or part-time	2/21/2018	£15,000 - £17,000 per annum, pro-rata	West Sussex
Permanent, full-time	3/1/2018	£37,000 - £45,000 per annum	South East England
Permanent, part-time	3/12/2018	£9,100 - £9,152 per annum, OTE	County Antrim
Permanent, full-time	2/28/2018	£9.25 - £10.15 per hour	Norfolk
Permanent, full-time	3/2/2018	£35,000 - £40,000 per annum	Bedfordshire
Permanent, full-time	2/28/2018	£23,000 - £26,600 per annum	Lancashire
Permanent, full-time	3/12/2018	£18,000 - £25,000 per annum, inc benefits	West Yorkshire
Permanent, full-time	3/12/2018	£12,000 - £40,000 per annum, OTE	Warwickshire
Permanent, full-time	3/12/2018	£30,000 - £40,000 per annum	West Yorkshire
Permanent, full-time	3/6/2018	Competitive salary	Hertfordshire
Permanent, full-time	2/28/2018	Competitive salary	South East England

Showing 1 to 14 of 50,000 entries, 12 total columns

Data Clean and Inspect

Preprocessing the Dataset and Handling Missing Values

Purpose

- Ensure data consistency and completeness before analysis.
 - Remove or handle null, incorrect, or irrelevant entries.
-

Steps Performed

- Checked and counted missing values across all columns
 - Removed rows with missing values in critical fields: job_title, city, and salary_offered
 - Replaced NA values in job_type and category columns with "Unknown"
 - Verified column renaming (e.g., job_tittle → job_title) to ensure schema consistency
-

Result

- Dataset cleaned and reduced to include only valid, analyzable rows

Ensured no major fields are left with NA values for the remainder of the analysis

Code Snippet

```
39 # =====
40 # STEP 3: Clean and Inspect
41 # =====
42 # Clean column names
43 jobs <- clean_names(jobs)
44
45 # Check structure of data
46 str(jobs)
47
48 # Summarize with skim
49 skim(jobs)
50 |
51 # View distinct values for key columns
52 cat("\nDistinct job types:\n")
53 print(unique(jobs$job_type))
54 cat("\nDistinct categories:\n")
55 print(unique(jobs$category))
56 cat("\nDistinct states:\n")
57 print(unique(jobs$state))
58
```

Data	
data	50000 obs. of 12 variables
job_in_uk	50000 obs. of 12 variables

Handle Missing Values

Managing Null Entries for Data Integrity

Purpose

- Identify and handle incomplete or inconsistent data
 - Ensure no analytical distortion caused by missing fields
-

Approach Used

- Used `colSums(is.na())` to quantify missing data in each column
 - Dropped rows with NA values in critical fields: `job_title`, `city`, `salary_offered`
 - Imputed Unknown for NA values in `job_type` and `category`
 - Verified that important columns are complete post-cleaning
-

Outcome

- Dataset refined to contain only rows with valid and analyzable information
 - Critical columns are free of nulls, ensuring better modeling and visualization later on
-

Code Snippet

```
# =====  
# STEP 4: Handle Missing Values  
# =====  
# Check missing value counts  
cat("\nMissing values per column:\n")  
print(colSums(is.na(jobs)))  
  
# Rename incorrect column if necessary  
if ("job_titile" %in% names(jobs)) {  
  jobs <- rename(jobs, job_title = job_titile)  
}  
  
# Remove rows with key missing data  
jobs <- drop_na(jobs, job_title, city, salary_offered)  
  
# Fill missing job_type and category with "Unknown"  
jobs$job_type[is.na(jobs$job_type)] <- "Unknown"  
jobs$category[is.na(jobs$category)] <- "Unknown"
```

Data Cleaning

Preparing the Dataset for Reliable Analysis

Purpose

- Standardize and format key fields for consistency and accurate interpretation
 - Remove anomalies and irrelevant variations before visualization
-

Steps Performed

- Parsed `post_date` using `dmy()` to convert to standard date format
 - Cleaned `salary_offered` by removing non-numeric characters and converting to numeric
 - Standardized categorical fields using `str_to_title()` for uniform capitalization
 - Removed salary outliers above the 99th percentile to avoid skewing distributions
-

Outcome

- Uniform and clean textual data enables easier grouping and analysis
 - Salary field standardized for numerical comparison
 - Irregular outlier values removed for clearer visual insights
-



Code Snippet

```
# =====  
# STEP 5: Data Cleaning  
# =====  
# Convert dates properly  
jobs$post_date <- dmy(jobs$post_date)  
  
# Clean and convert salary field  
jobs$salary_offered <- gsub("[^0-9]", "", jobs$salary_offered)  
jobs$salary_offered <- as.numeric(jobs$salary_offered)  
  
# Standardize text fields  
jobs$job_title <- str_to_title(jobs$job_title)  
jobs$category <- str_to_title(jobs$category)  
jobs$job_type <- str_to_title(jobs$job_type)  
jobs$state <- str_to_title(jobs$state)  
  
# Remove outlier salaries beyond 99th percentile  
salary_threshold <- quantile(jobs$salary_offered, 0.99, na.rm = TRUE)  
jobs <- jobs |>  
  filter(salary_offered <= salary_threshold)
```

Data Visualization & Exploration

Uncovering Patterns and Trends in the UK Job Market

Purpose

- Gain insights into job distribution, demand, and salary trends
 - Use visualizations to interpret data across categories, locations, and time periods
-

Visualizations Created

-  **Top 10 Job Titles:**
 - Bar chart of most frequently posted job titles
-  **Top 10 Cities by Job Count:**
 - Visualizing regions with highest employment opportunities
-  **Salary Distribution:**
 - Histogram showing range and frequency of offered salaries
-  **Job Postings Over Time:**
 - Line chart tracking monthly posting trends
-  **Job Types Distribution:**
 - Frequency of full-time, part-time, contract jobs, etc.

-  **Top Job Categories:**
 - Identifying dominant employment sectors
 -  **Top Hiring Companies:**
 - Companies with the most job listings
-

✓ Insights Gained

- Data science and tech jobs highly listed
 - London and Manchester lead in job availability
 - Salary trends concentrated within a predictable range
 - Full-time roles dominate the listings
-

Code Snippet

```
# =====  
# STEP 6: Visualizations  
# =====  
  
jobs |>  
  count(job_title, sort = TRUE) |>  
  slice_max(n, n = 10) |>  
  ggplot(aes(x = reorder(job_title, n), y = n)) +  
  geom_col(fill = "#2E86AB") +  
  coord_flip() +  
  labs(title = "Top 10 Job Titles in UK", x = "Job Title", y = "Count") +  
  theme_minimal()
```

```
jobs |>
  count(city, sort = TRUE) |>
  slice_max(n, n = 10) |>
  ggplot(aes(x = reorder(city, n), y = n)) +
  geom_col(fill = "#FF8C42") +
  coord_flip() +
  labs(title = "Top 10 Cities by Job Count", x = "City", y = "Jobs") +
  theme_minimal()
```

```
jobs |>
  ggplot(aes(x = salary_offered)) +
  geom_histogram(bins = 30, fill = "#28B463", color = "white") +
  labs(title = "Salary Distribution", x = "Salary", y = "Frequency") +
  xlim(0, quantile(jobs$salary_offered, 0.95, na.rm = TRUE)) +
  theme_minimal()
```

```
jobs |>
  filter(!is.na(post_date)) |>
  mutate(month = floor_date(post_date, "month")) |>
  count(month) |>
  ggplot(aes(x = month, y = n)) +
  geom_line(color = "#8E44AD", size = 1.2) +
  labs(title = "Job Postings Over Time", x = "Month", y = "Number of Jobs") +
  theme_minimal()
```

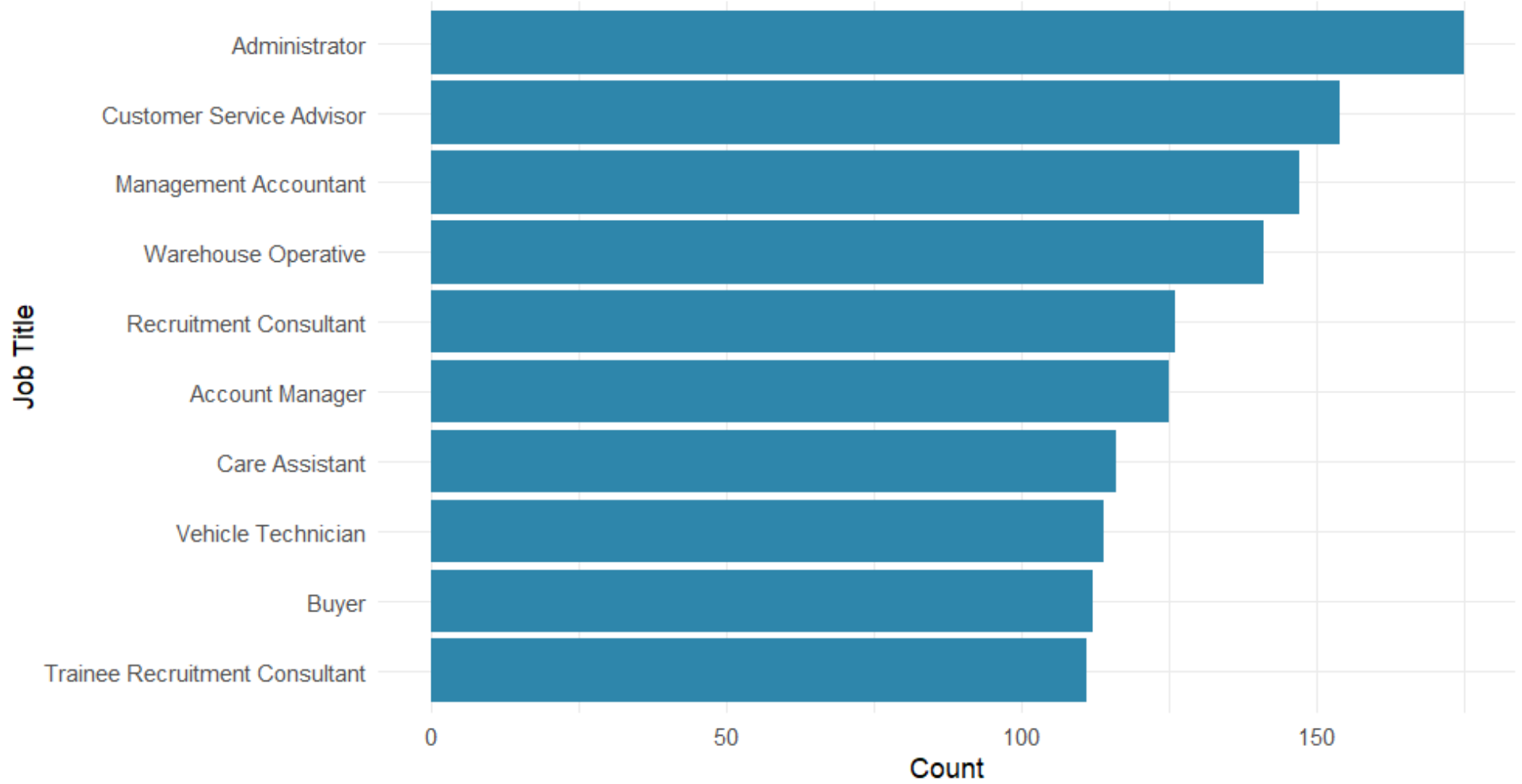


```
jobs |>
  count(job_type, sort = TRUE) |>
  ggplot(aes(x = reorder(job_type, n), y = n, fill = job_type)) +
  geom_col() +
  coord_flip() +
  scale_fill_brewer(palette = "Set2") +
  labs(title = "Job Types Distribution", x = "Job Type", y = "Count") +
  theme_minimal() +
  theme(legend.position = "none")

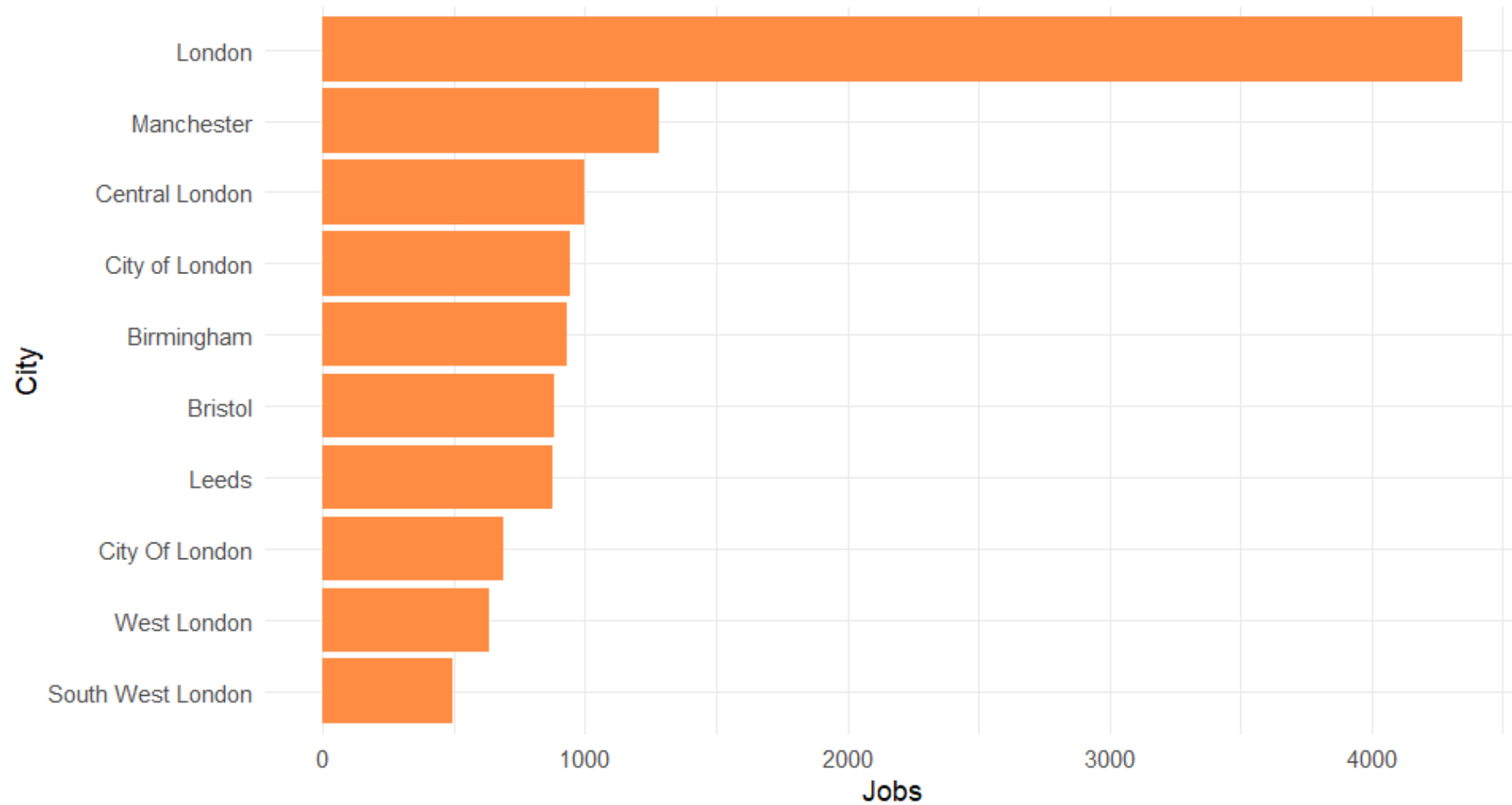
jobs |>
  count(category, sort = TRUE) |>
  slice_max(n, n = 10) |>
  ggplot(aes(x = reorder(category, n), y = n)) +
  geom_col(fill = "#3498DB") +
  coord_flip() +
  labs(title = "Top Job Categories", x = "Category", y = "Number of Jobs") +
  theme_minimal()

jobs |>
  count(company_name, sort = TRUE) |>
  slice_max(n, n = 10) |>
  ggplot(aes(x = reorder(company_name, n), y = n)) +
  geom_col(fill = "#E67E22") +
  coord_flip() +
  labs(title = "Top Companies Hiring", x = "Company", y = "Job Count") +
  theme_minimal()
```

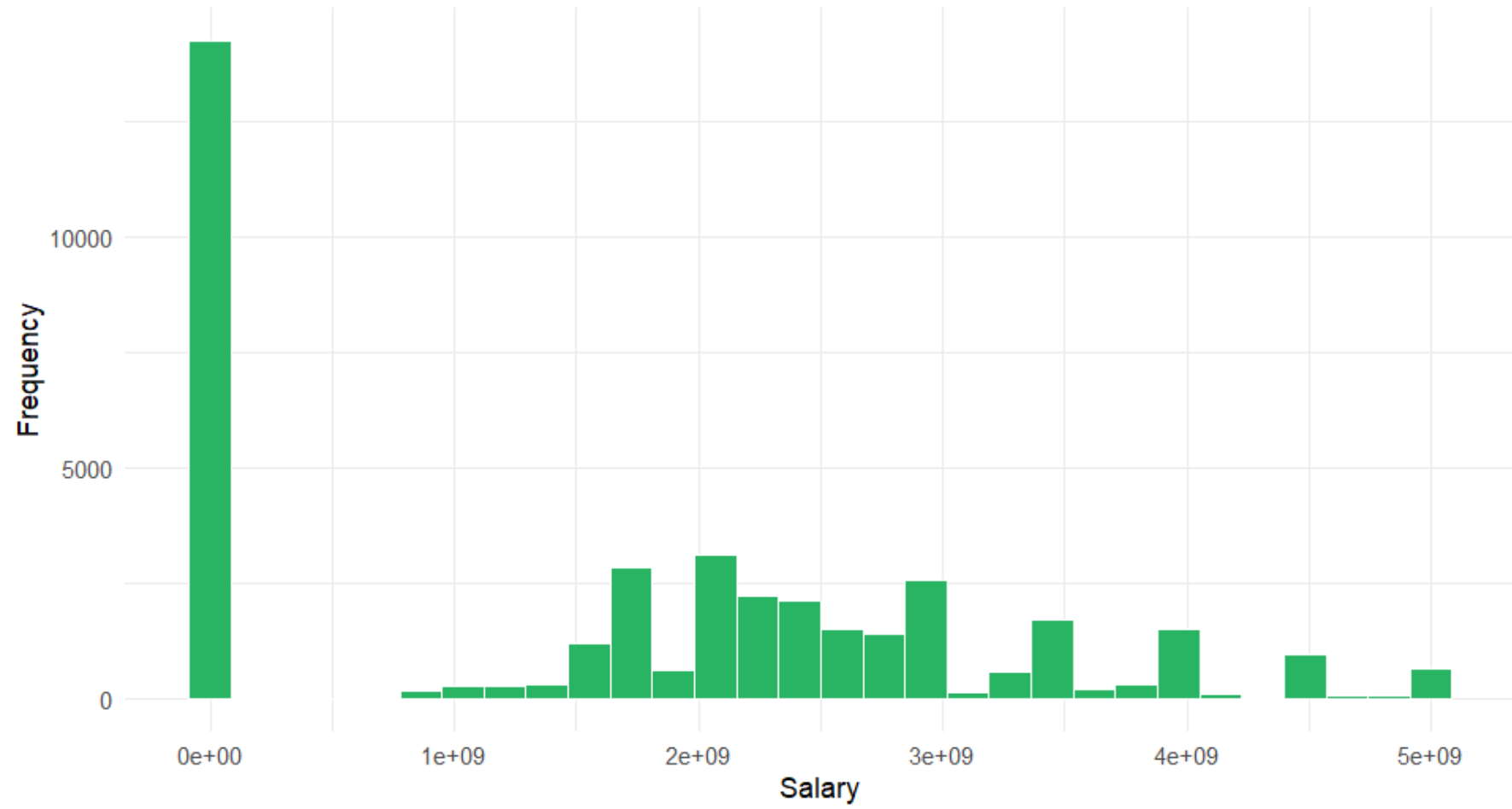
Top 10 Job Titles in UK



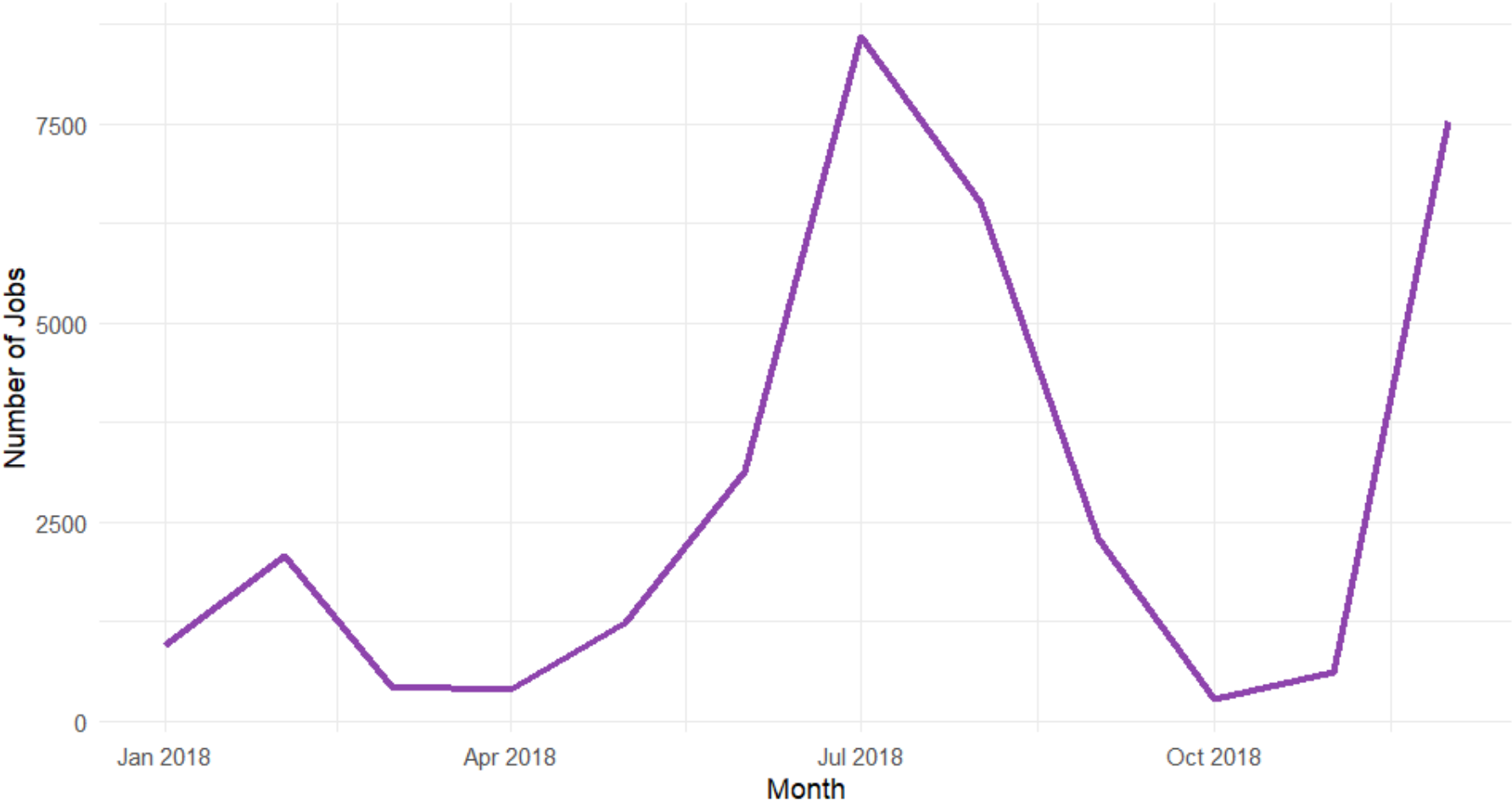
Top 10 Cities by Job Count



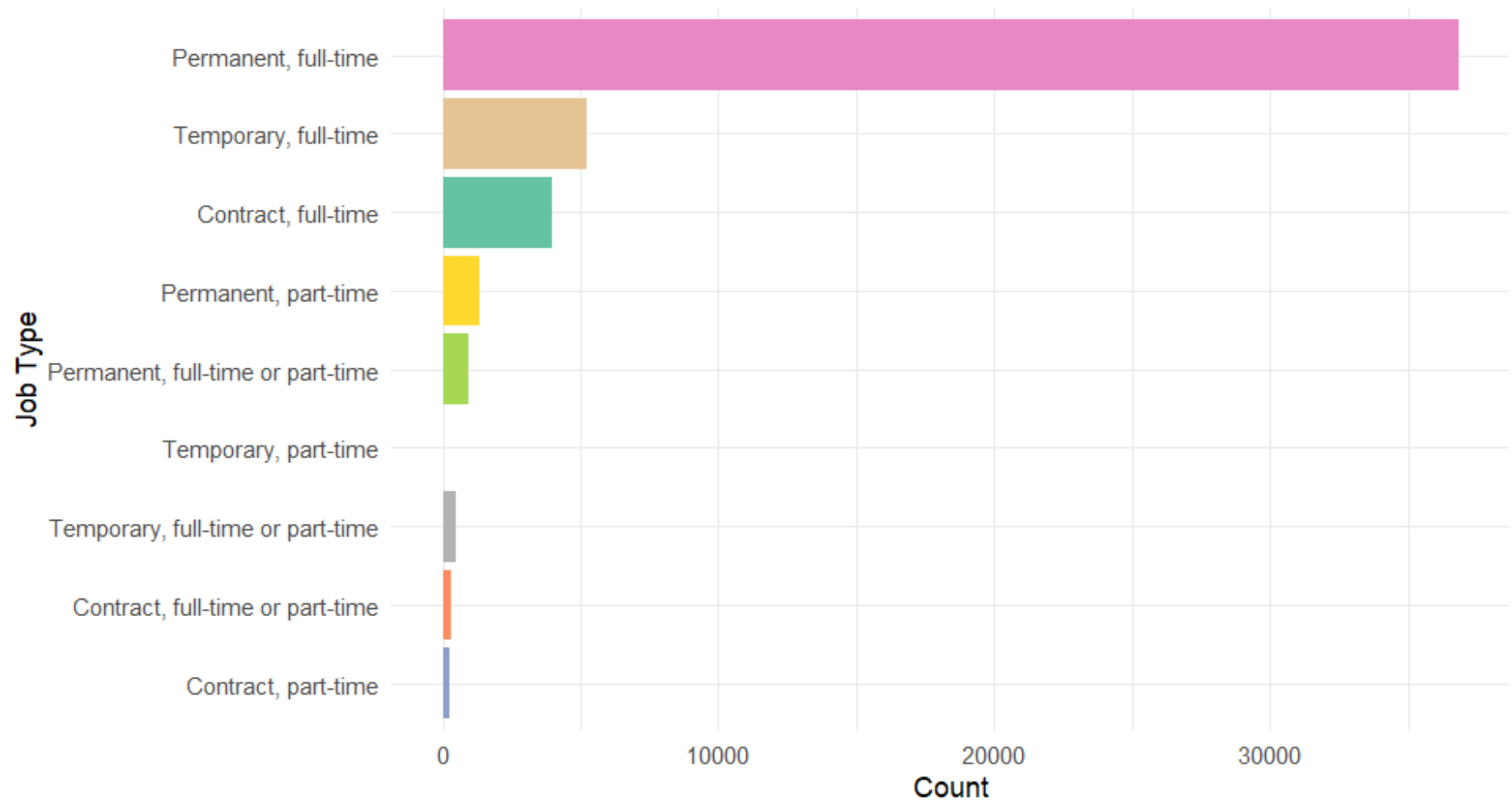
Salary Distribution



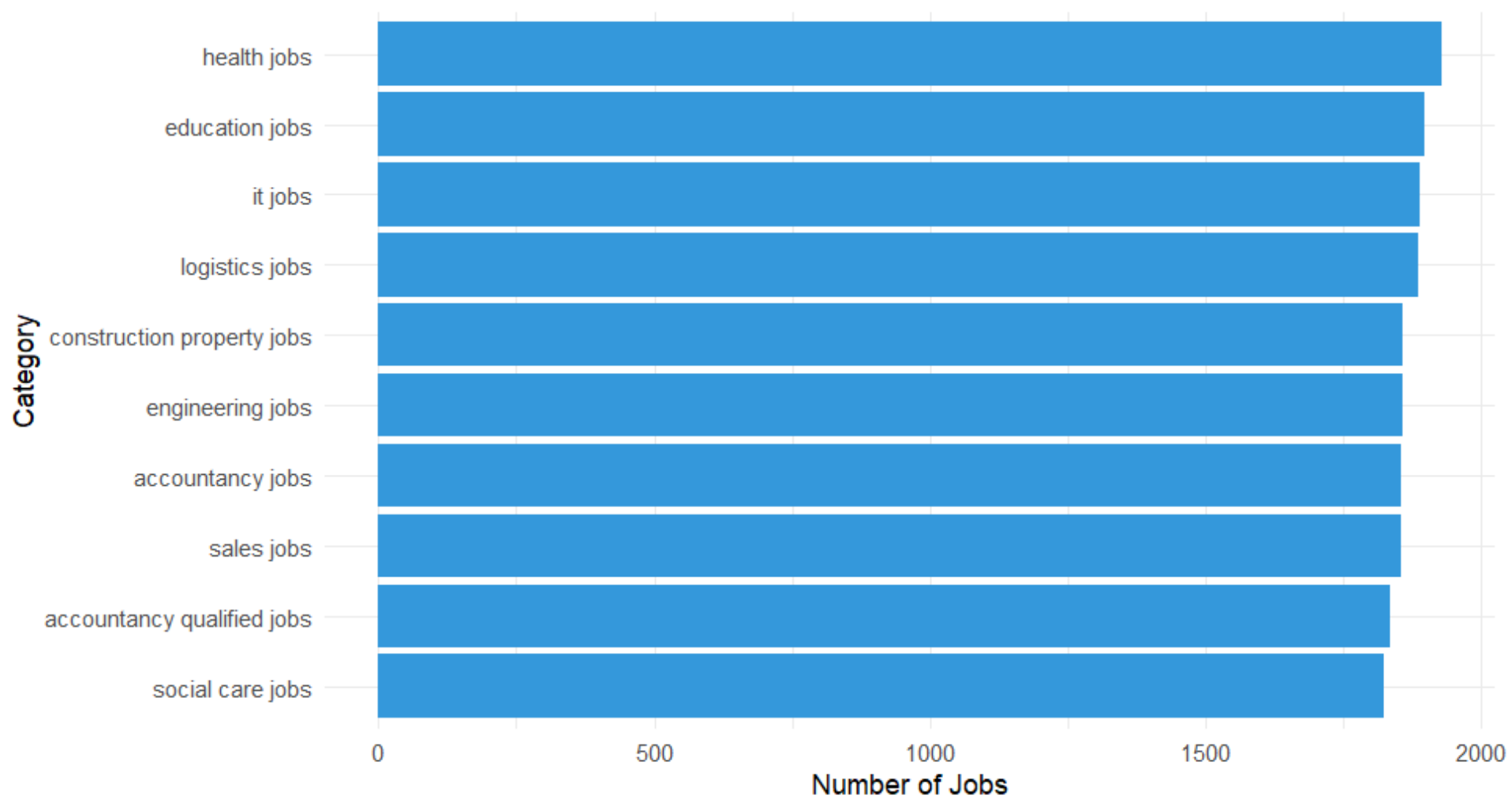
Job Postings Over Time



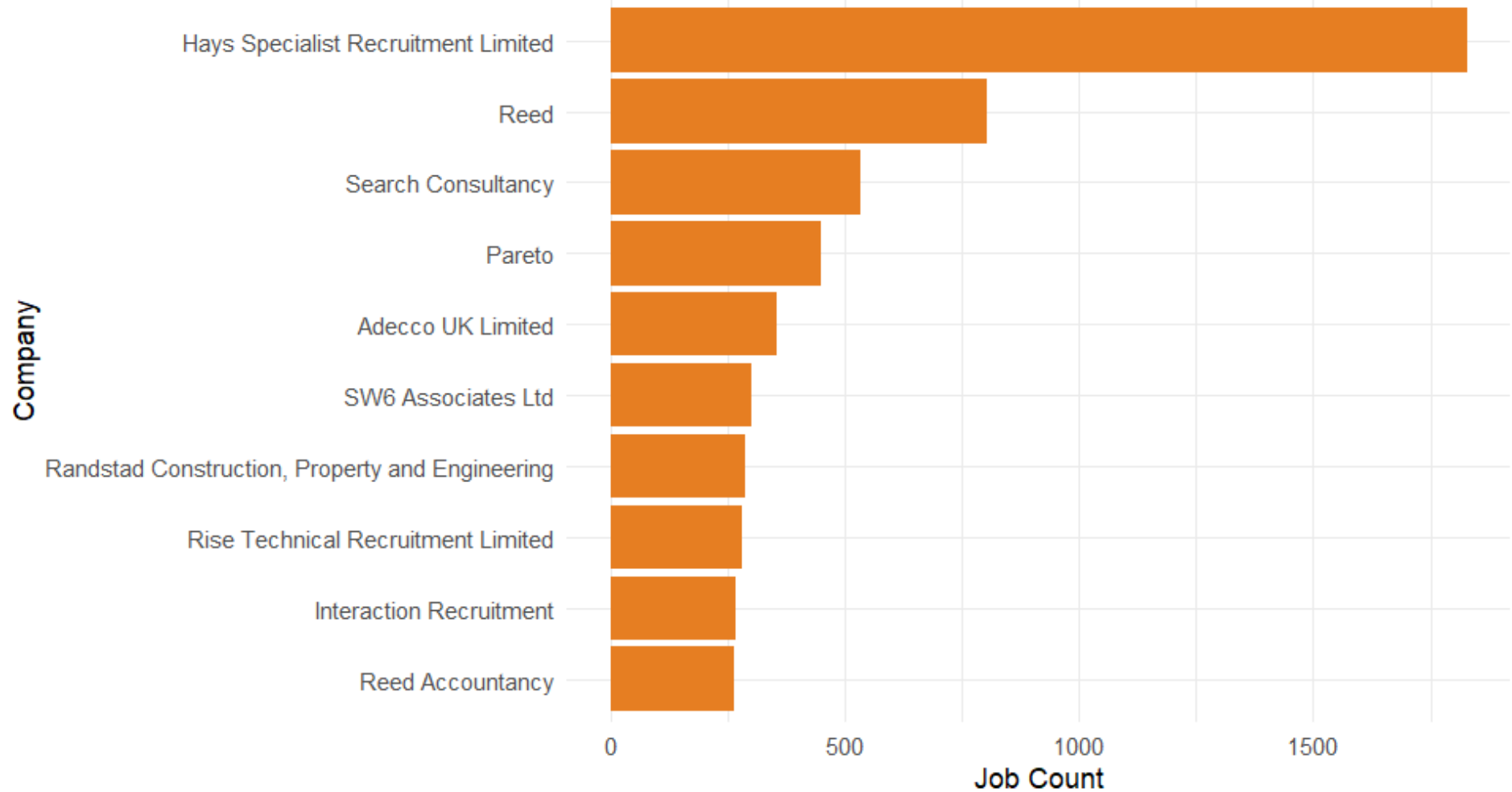
Job Types Distribution



Top Job Categories



Top Companies Hiring



Text Analysis on Job Descriptions

Extracting Keywords from Job Postings Using NLP

Purpose

- Identify the most frequent and relevant words from job descriptions
 - Uncover skill trends and demand from textual content
-

Steps Performed

- Used `unnest_tokens()` to break job descriptions into words
- Removed stopwords using the stopwords package
- Counted word frequencies and identified the top 20 keywords
- Created a bar chart and optionally a word cloud for visualization

Insights Gained

- Frequently appearing terms reveal popular skills and job responsibilities
 - Word frequency reflects hiring focus across sectors and roles
-

Code Snippet

```
# =====  
# STEP 7: Text Analysis on Job Descriptions  
# =====  
  
cat("\nAnalyzing the most common words in job descriptions...\n")  
jobs_words <- jobs |>  
  unnest_tokens(word, job_description) |>  
  anti_join(get_stopwords()) |>  
  count(word, sort = TRUE)  
  
cat("\nTop 20 most common words in job descriptions:\n")  
print(head(jobs_words, 20))  
  
jobs_words |>  
  slice_max(n, n = 20) |>  
  ggplot(aes(x = reorder(word, n), y = n)) +  
  geom_col(fill = "#5DADE2") +  
  coord_flip() +  
  labs(title = "Most Common Words in Job Descriptions", x = "Word", y = "Frequency") +  
  theme_minimal()  
  
response <- readline(prompt = "\nWould you like to generate a word cloud of job descriptions? (yes/no): ")  
if(tolower(response) == "yes") {  
  wordcloud(words = jobs_words$word, freq = jobs_words$n, min.freq = 20,  
            max.words = 100, random.order = FALSE, colors = brewer.pal(8, "Dark2"))  
}
```

leading knowledge days services environment
consultant required manager customers
level excellent management good
across contact
day support business well people
strong based skills recruitment able
part working apply sales ability
need please work a role client career
key job us care data new position
training experience high salary
also can customer team time must
benefits join staff within full offer company
one office ensure looking service development quality clients agency
opportunities opportunity including employment include
project provide successful candidate hours

Salary by Job Category

Comparing Average Salaries Across Different Sectors

Purpose

- Understand how salaries vary by job category
 - Identify which sectors offer the highest average compensation
-

Steps Performed

- Grouped the data by category
 - Calculated the average salary for each category
 - Visualized the top 10 categories with the highest average salaries using a bar chart
-

Insights Gained

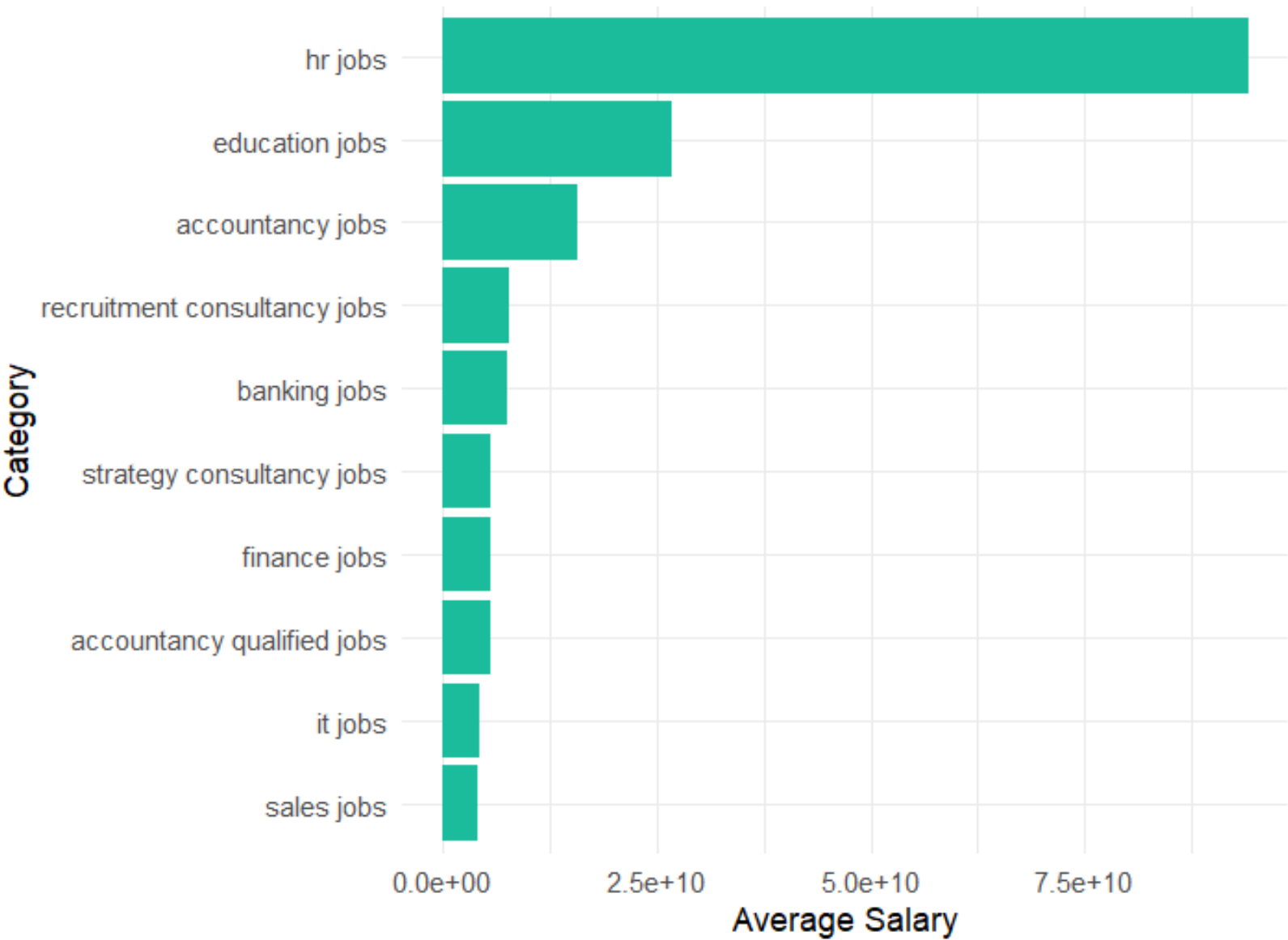
- Certain industries like Tech, Finance, and Engineering lead in salary offerings
 - Visual comparison helps prioritize sectors based on financial potential
-



Code Snippet

```
# =====  
# STEP 8: Salary by Job Category  
# =====  
jobs |>  
  group_by(category) |>  
  summarise(avg_salary = mean(salary_offered, na.rm = TRUE)) |>  
  slice_max(avg_salary, n = 10) |>  
  ggplot(aes(x = reorder(category, avg_salary), y = avg_salary)) +  
  geom_col(fill = "#1ABC9C") +  
  coord_flip() +  
  labs(title = "Average Salary by Job Category", x = "Category", y = "Average Salary") +  
  theme_minimal()
```

Average Salary by Job Category



Jobs by State

Regional Distribution of Job Listings in the UK

Purpose

- Analyze the geographic spread of job opportunities across UK states
 - Identify regional employment hubs
-

Steps Performed

- Grouped job data by state
 - Counted job listings per state
 - Selected the top 10 states with the highest job counts
 - Visualized using a horizontal bar chart
-

Insights Gained

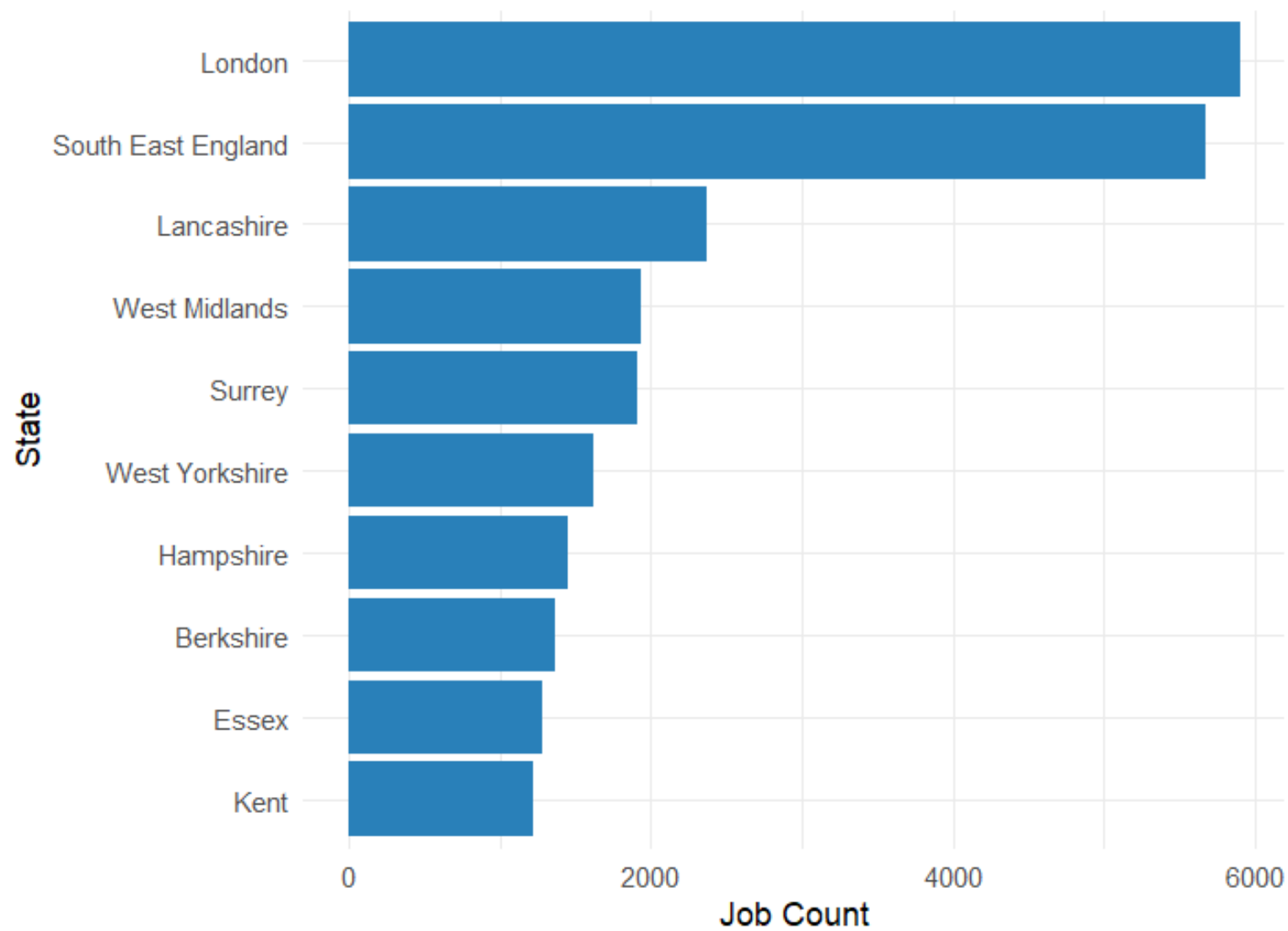
- Certain states dominate in job postings (e.g., England)
 - Helps in identifying high-demand regions for employment
-



Code Snippet

```
# =====  
# STEP 9: Jobs by State  
# =====  
jobs |>  
  count(state, sort = TRUE) |>  
  slice_max(n, n = 10) |>  
  ggplot(aes(x = reorder(state, n), y = n)) +  
  geom_col(fill = "#2980B9") +  
  coord_flip() +  
  labs(title = "Jobs by State", x = "State", y = "Job Count") +  
  theme_minimal()
```


Jobs by State



Job Search by Field and State

Interactive Filtering Based on User Preference

Purpose

- Provide a user-friendly way to filter jobs by both industry and region
 - Allow interactive exploration of job opportunities based on interest
-

Steps Performed

- Prompted the user to enter a keyword related to job title (e.g., Data, Engineer)
 - Prompted the user to specify a state (e.g., England, Scotland)
 - Used `str_detect()` to match user input with job titles and states
 - Displayed top job listings if a match was found; otherwise, showed a warning
-

Insights Gained

- Enables real-time job filtering tailored to user preferences
 - Facilitates personalized job exploration by field and region
-



Code Snippet

```
# =====
# STEP 10: Job Search by Field and State
# =====
|
# Ensure job_title and state fields are not NA and are character
jobs <- jobs %>%
  filter(!is.na(job_title), !is.na(state)) %>%
  mutate(
    job_title = as.character(job_title),
    state = as.character(state)
  )

cat("\nSearch for Jobs Based on Your Interest!\n")
field_input <- readline(prompt = "Enter a job field (e.g., Data, Engineer, Marketing): ")
state_input <- readline(prompt = "Enter a state (e.g., England, Scotland): ")

if (field_input == "" | state_input == "") {
  cat("\nPlease enter both field and state to proceed with the search.\n")
} else {
  filtered_jobs <- jobs %>%
    filter(str_detect(tolower(job_title), tolower(field_input)),
           str_detect(tolower(state), tolower(state_input)))

  if (nrow(filtered_jobs) == 0) {
    cat("\nNo jobs found for that combination. Try another field or state.\n")
  } else {
    cat("\nHere are some job titles matching your search:\n")
    print(
      head(
        filtered_jobs %>%
          select(job_title, company_name, city, salary_offered),
        10
      )
    )
  }
}
```

Conclusion

Key Takeaways from the UK Jobs Data Analysis

Project Summary

- Conducted a comprehensive analysis of UK job listings using real-world data from Reed.co.uk
 - Cleaned and standardized the dataset for accurate analysis
 - Uncovered trends through visualizations, such as top job titles, hiring companies, salary ranges, and regional demand
 - Performed text mining to identify skill demand via job description keywords
 - Enabled interactive job filtering by field and state
-

Key Insights

- Tech and business roles dominate the job market
 - Major cities like London and Manchester host the majority of job opportunities
 - Full-time jobs are more prevalent than part-time or contractual roles
 - Average salary varies significantly by job category, with tech-related roles often leading
-

Future Scope

- Integrate machine learning to predict job trends or salary estimations
- Include real-time job data updates for continuous analysis